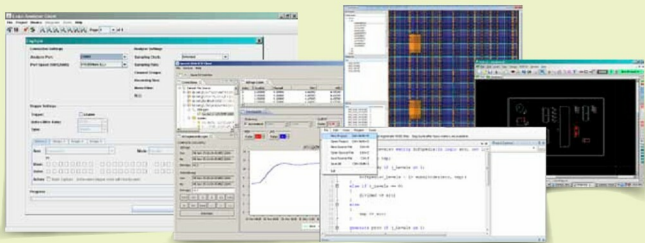


Electronics Development Tools

In this month's EFY Plus DVD, you can look forward to some excellent software for chip design as well as digital design. You will also find some really effective SCADA and PCB design software



SRINIVAS N.C.

SynaptiCAD

SynaptiCAD design suite aims to overcome the disadvantage of traditional tools where users have to almost complete a design before being able to get simulation results. With SynaptiCAD tools, they can begin analysing and simulating design ideas right away without having to complete an entire circuit model or schematic. The tools automatically handle many of the minor details involved in the design effort, leaving them more time to focus on the bigger picture.

The design suite consists of a number of tools such as VeriLogger Extreme, BugHunter, TimingDiagrammer, WaveFormer and TestBench. Each of these tools provides a separate functionality to the user. The software comes with an extensive documenta-

tion and tutorials that allow new users to get familiar with the tools quite easily.

BugHunter uses SynaptiCAD graphical environment and supports all major HDL simulators. It can launch the simulator and provide single-step debugging. It also features unit-level test bench generation, streaming of waveform data, project management and a hierarchy tree.

The unit-level test bench generation is unique in that it lets the user draw stimulus waveforms, then generate the stimulus model and wrapper code, and launch the code. With an integrated debugging environment you can graphically build a project, launch simulation and view the results in just a few minutes. The interface also manages the test bench interface so that it is easy to create a set of regression tests to run the design through.

Timing Diagrammer consists of a modeless drawing and editing en-

vironment, delays, setups and holds for performing timing analysis, time markers, seven graphical waveform states, virtual and group buses, clocks with formulae.

WaveFormer is a rapid-prototyping EDA tool that helps you design faster with fewer mistakes. It enables you to automatically determine critical paths, verify timing margins, adjust for re-convergent fan-out effects, and perform "what if" analysis to determine optimal clock speed. It also lets you specify and analyse system timing and perform Boolean level simulation without the need for schematics or simulation models. When the timing diagram is complete, you can generate digital stimuli for your favourite Verilog, VHDL, SPICE or gate-level simulator.

TestBench is a graphical test bench generator that greatly reduces the time required to create and maintain test benches. One of the most